MIL-P-2497B(MU)
11 SEPTEMBER 1968
SUPERSEDING
MIL-P-2497A(ORD)
6 OCTOBER 1961

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M27 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Percussion, M27.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for.

MIL-I-45208 — Inspection Requirements; General Specification for.

MIL-I-45607 — Inspection Equipment; Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8837810 — Primer, Percussion, M27, Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-8837810 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer. The primer shall comply with all requirements specified on Drawing (dwg.) 8837810 and with all requirements specified in applicable specifications.

3.3 Functioning.

3.3.1 Sensitivity (see 6.4). The primer shall function under an applied energy of 23.52 inch ounces maximum (max.) (see 6.5) without evidence of squibbing or hangfire, but shall not function under an applied energy of 3.84 inch

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ounces minimum (min.) when tested as specified in 4.3.1.1.

3.3.2 *Duds.* The primers shall fire without any indication of a failure (see 6.6), when tested as specified in 4.3.1.2.

3.4 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208, except that height of drop and weight of ball for the sensitivity functioning test shall not vary from that specified in 4.3.1.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

(b) Number of units of product inspected.

(c) Results obtained for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The intitial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required wih subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45308 and will be performed to the extent necessary to assure compliance with the contractual requirement. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of primers produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specifica-

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tion, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Each lot shall contain:

(a) Parts bearing the same lot interfix number from one manufacturer.

(b) Primer mixture from one batch or blending of batches which results in homogeniety of all primer mixture used in loading the entire primer lot.

4.2.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Loaded cup (see dwg. 8837810).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge missing	Visual	01001
102.	Foil paper cover missing	Visual	01002
103.	Cup distorted	Visual	01003
104.	Primer mixture on exterior	Visual	01004
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	01005

4.2.2.2 Assembly (see dwg. 8837810).

Categories	Defects	Method of inspection	Code No.
Critical:			
1.	Crack or split	Visual	02001
Major:	AQL 0.65 percent		
101.	Total length	Gage	02002
102.	Depth to cup	Gage	02003
103.	Outside diameter, minimum (min.)	Gage	02004
104.	Anvil missing	Visual	02005
105.	Hole in body obstructed	Visual	02006
106.	Assembly damaged	Visual	02007
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship except as defined in Critical 1 (see 3.4)	Visual	02008

4.2.3 Testing.

4.2.3.1 Sensitivity (see 3.3.1) Major defect Code No. 02009. A sample of 550 percussion primers shall be selected from each lot for this test. If the sum of the average critical height (H average) plus 5 times the standard deviation (s) multiplied by the ball weight exceeds the maximum requirement, or if the sum of the average critical height (H average) minus 2 times the standard deviation (s) multiplied by the weight of the ball is less than the minimum requirement, the lot shall be rejected (see 6.4). The test shall be performed as specified in 4.3.1.1, using test equipment specified in 4.2.4.

4.2.3.2 *Duds (see 3.3.2), Major defect—Code No. 02010.* The sampling plan for this test shall be in accordance with Standard MIL-STD-105, Table IV A, using code letter N and an AQL of 0.10 percent. The test shall be performed as specified in 4.3.1.2, using test equipment specified in 4.2.4.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL-8837810 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment. The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Functioning.* Prior to performing this test, the primers shall be conditioned at a temperature of 75° (F.) plus or minus 5° F. for a period of 24 hours, minimum.

4.3.1.1 *Sensitivity.* The sample of 550 primers shall be divided into 11 groups of 50 primers each and tested in the following manner:

The first group (50 primers) of primers shall be tested by dropping a steel ball $\frac{1}{4}$ inch in diameter and weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 7 inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball.) The number of primers that fire, fail to fire, hangfire or squib (see 6.6) shall be recorded. This procedure shall then be repeated at successively increasing heights of 1 inch using a new group of 50 primers at each different height, until a height is reached at which all primers in the group being tested fire. The test fixture shall then be set at a height of 6 inches and the above procedure repeated at successively decreasing heights of 1 inch, using new groups of 50 primers at each height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection shall be determined from calculations based on recorded data (see 6.4). All samples subjected to test shall be scrapped. Any remaining untested samples may be returned to the lot from which they are taken.

4.3.1.2 *Duds.* Each primer of the sample of 300 primers shall be tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of "H average" plus 3 "s." In no case will this height exceed 12 inches. The value of "H average" plus 3 "s" shall be calculated from data obtained in 4.3.1.1 using procedures outlined in 6.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The primers shall be preserved, packaged and packed in standard commercial containers so as to assure acceptance by common or other carrier for safe transportation, in accordance with Interstate Commerce Commission Regulations, at the lowest rate to the point of delivery. The primers shall be packed and spaced in such manner that a barrier of some material is present between the crown of one primer and the base of another primer in order to resist mass detonation and shield the primers from stray electrical current, flame and shock.

5.2 Marking. Packing boxes containing the primers shall be marked in accordance with Standard MIL-STD-129.

5.3 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following.

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection Equipment Lists.* Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the

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"number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection Equipment List codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF — Contractor's design responsibility on Ordnance format; ORDM–608–12.

CDCF — Contractor's design responsibility on contractor format.

OD — Ordnance design.

ODMU — Ordnance design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and.

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in addition, contain the detail and information specified above.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Formula and Method for determining H average and s.

6.4.1 *Formula:*

H average = A plus B plus C.

s = I times the square root of L minus M.

where:

A = the sum of column 3.

B = the first drop height at which all elements misfired times 100 percent.

C = one half the drop height increment (see 6.7).

L = the sum of column 5.

M = the square of the sum of column 3.

I = the drop height increment.

6.4.2 *Tabulation of data.* Divide a sheet of paper into 5 columns and tabulate the data as follows:

(a) Title column 1 "Height of Drop" and list all drop heights in consecutive order starting with the lowest height. DO NOT list the heights at which all elements fired or misfired.

(b) Title column 2 "Number Misfired" and list the sum of the number of elements that misfired and "squibed" (see 6.6), at each intermediate height.

(c) Title column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that misfired at each intermediate height. This number is obtained by dividing the value obtained under "Number Misfired" by the number of elements tested at that height. Calculation should be carried out and recorded to the nearest one-hundredth (.01).

(d) Title column 4 "Standard Deviation Factor" and list all numerals in sequence starting with 1, for as many drop heights as are listed in column 1.

(e) Title column 5 "Standard Deviation Factor Times Fraction Misfired" and, by multiplying each decimal fraction in Column 3 by the number on the same line in column 4, enter the product on the corresponding line in column 5.

6.4.3 *Procedure for solving equations.*

(a) To obtain "A"—add the values in column 3 and substitute the sum for "A."

(b) To obtain "B"—multiply the first drop height at which all elements misfired by 100 percent and substitute the product for "B."

(c) To obtain "C"—divide the drop height increment by 2 and substitute the dividend for "C."

(d) To obtain "L"—add the values in column 5 and substitute the sum for "L."

(e) To obtain "M"—square the sum of column 3 and substitute the sum for "M."

(f) To obtain "H Average"—add the values substituted for "A, B, and C."

(g) To obtain "S"—Subtract the value substituted for "M" from the value substituted "L" and then obtain the square root by the root of the difference, and then multiply the square root by the height of increment.

(h) Using the calculated values for "H Average" and "s," if "H Average" plus 5 "s" multiplied by the maximum weight of the ball is equal or less than the maximum inch ounce energy value required, and "H Average" minus 2 "s" multiplied by the minimum weight of the ball is equal or greater than the minimum energy value required, the lot should be considered acceptable for sensitivity.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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